Patrick Loiseau

Research Director at Inria (Saclay) — head of FairPlay team
Professor (part-time) at École Polytechnique (DIX) and at ENSAE (responsable for CS courses)

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Research Interests

My research revolves around *machine learning/AI* and *game theory*, with particular interest on online learning and algorithms, economics of data (incentives, data valuation, data markets), trustworthy machine learning (fairness-aware algorithms, security of agentic AI, LLM alignment), and applications (matching, online advertising).

Education

Université Pierre et Marie Curie (Paris 6) — Paris, France
HDR (habilitation à diriger des recherches) — Dec. 2016
Thesis: *Combining game theory and statistical learning for security, privacy and network systems*

Université Pierre et Marie Curie (Paris 6) / École Polytechnique — Paris, France
M.Sc. in Mathematics – *Probability and random models* (mention très bien) — July 2010
Thesis: *Large deviations for mixing processes*

École Normale Supérieure de Lyon (LIP lab) — Lyon, France
Ph.D. in Computer Science (advisors: Paulo Gonçalves, Pascale Vicat-Blanc Primet) — Dec. 2009
Thesis: *Contributions to the Analysis of Scaling Laws and Quality of Service in Networks: Experimental and Theoretical Aspects*

École Normale Supérieure de Lyon — Lyon, France
"Elève normalien" (BSc, agrégation, MSc) in Physics — 2002 – 2006

- M.Sc. in Physics – *Non linear and statistical physics* (July 2006)
  Thesis: *Complex wavelets for the analysis of scaling phenomena*

Professional Experience

Inria — Grenoble then Saclay, France
Research director (DR) — since Oct. 2024
Research scientist (CRCN) — Nov. 2018 – Sept. 2024

- Head (with V. Perchet) of the FairPlay team in Saclay: a joint team Criteo-ENSEA-Inria with 8 researchers and 20 PhD students/postdocs that we created (since Mar. 2022).
  Member of the POLARIS team (Inria-LIG) in Grenoble (until Feb. 2022)
- Co-holder (with V. Perchet, Y. Tamura, P. Winant) of a **Hi! Paris** synergy fellowship on *Design, Incentivization, Optimization and (Reinforcement-)Learning of Multi-Layered Market* (since 2025)
- Co-holder (with M.-C. Rousset) of a Chair of **MIAI @ Grenoble Alpes** on *Explainable and Responsible AI* (2019 – 2023)

ENSAE — Palaiseau, France
Part-time Professor, responsible for the CS courses — since Sept. 2022

École Polytechnique — Palaiseau, France
Part-time Professor in the CS department (DIX) — since Sept. 2020

Univ. Grenoble Alpes Grenoble, France
Holder of a Chair of Excellence IDEX UGA at the LIG lab. Oct. 2017 – Oct. 2018

EURECOM Sophia-Antipolis, France
Assistant Professor in the Data Science department Nov. 2011 – Aug. 2017

University of California, Santa Cruz Santa Cruz, CA, USA
Post-doctoral scholar in Basking Engineering school Dec. 2010 – Oct. 2011

Inria Paris-Rocquencourt Le Chesnay, France
Post-doctoral fellow in Sisyphe team Jan. 2010 – Nov. 2010

Visiting positions

Max Planck Institute for Software Systems Saarbrücken, Germany
Visiting researcher (*Humboldt Research Fellowship for experienced researchers*) April 2016 – Oct. 2018
Visiting researcher July 2014 – Sept. 2014

University of California, Berkeley Berkeley, CA, USA
Visiting researcher in the EECS department July 2012 – Aug. 2012
Visiting member of the Network Economics Group Dec. 2010 – Oct. 2011

University of Waterloo Waterloo, ON, Canada
Visiting researcher in the ECE department Oct. 2010

Teaching experience

ENSAE Paris Palaiseau, France

Algorithm Design and Analysis Spring, Since 2023
Responsible of the teaching unit, lectures (18h) and TDs (6h), graduate level (Ingénieur 3A / MSc)

Projet de programmation ("Programming project") Winter, Since 2023
Responsible of the teaching unit, lectures/labs (18h) and organization of a programming project in Python, coordination of a team of 5 teaching assistants, undergraduate level (Ingénieur 1A)

École Polytechnique Palaiseau, France

Design and Analysis of Algorithms Winter, Since 2020
Petites classes (TD, 40h) and Projet Informatique (proposition and evaluation of a coding project for ∼15 groups of students), undergraduate level (Ingénieur 2A)

Advanced Machine Learning and Autonomous Agents Winter, Since 2020
Labs (TP, 12-16h) and lectures (2-4h), graduate level (Ingénieur 3A / M1)

Univ. Grenoble Alpes Grenoble, France

Introduction to Data Analysis Spring 2020
Co-responsible of the teaching unit, lectures (12h) and labs (TP, 9h), graduate level (M1 MOSIG)

INFO4: Probabilité et Simulation ("Probability and Simulation") Fall 2019
Lectures (6h) and labs (TD, 12h), graduate level (M1)

Algorithmes pour le traitement des données ("Algorithms for data processing") Fall 2019
Lectures (3h) and labs (TP, 12h), graduate level (M1)

Inf202: Modélisation des structures informatiques: aspects formels ("Formal aspects of computer structures") Spring 2018
Responsible for the teaching unit, lectures (15h) and tutorials (TD, 30h), coordination of a team of 6 teaching assistants, undergraduate level (L1)

EURECOM Sophia-Antipolis, France

Statistical data analysis Fall 2013–16
Instructor (new course), graduate course (short: 21 hours/year)

Game Theory Fall 2013–16
Instructor, graduate course (short: 21 hours/year)

Network Economics Fall 2012–16
Instructor (new course), graduate course (short: 21 hours/year)

Performance Evaluation of Computer Systems Spring 2012 and 2013
Instructor, graduate course (long: 42 hours/year)

University of California, Santa Cruz Santa Cruz, CA, USA

ISM207: *Random Process Models in Engineering* Spring 2011
Guest lecturer (3 hours), graduate course, TIM program

Professional service, scientific responsibilities and leadership

Teaching and internal responsibilities

Responsible for the computer science courses at ENSAE (since 2022)

Responsible for scientific monitoring of the ATOS-Inria collaboration around Grenoble (2020-21)

Responsible for the networking track of the engineering studies at EURECOM (2014-17)

Member of the restricted committee for reflection on EURECOM's long-term strategy (2014-16)

Steering committees

Chair of the steering committee of NetEcon (2013-21)

Creator and chair of the steering committee of Sophia-networking (2013-17)

Member of the scientific council of the Labex UCN@Sophia (2015-17)

Member of the steering committee of the "réseau thématique 2 (architecture et gestion)" of Institut Mines-Telecom (2014-17)

Conference organization

Member of the organizing committee of MATCH-UP 2026

Member of the organizing committee of Nash75 @Oxford (2025)

PC chair of the Hi! PARIS summer school "AI & Data for Science, Business and Society" in 2024 (with Shirish Srivastava) and 2025, then scientific committee member in 2026

Workshop chair of ACM SIGMETRICS 2021

Creator and lead organizer of the UCN@Sophia Labex seminar (a bimonthly seminar, 2013-17)

Member of the scientific programme committee of the BMW summer school 2017

Registration chair of ACM SIGMETRICS 2016

Co-organizer, seminar on modeling, optimization and control in wireless networks, Paris 2015

PC co-chair of NetEcon 2015 (with Aaron Roth and Adam Wierman)

PC co-chair of W-PIN+NetEcon 2014 (with John Chuang)

PC co-chair of W-PIN+NetEcon 2013 (with David Parkes and Jean Walrand)

Registration chair of ACM SIGMETRICS 2013

PC co-chair of W-PIN 2012 (with Jean Walrand)

Editorial activities

Associate editor, IEEE TBD (2018-21)

Associate editor, ACM TOIT (2015-21)

Guest editor, ACM TOIT special issue on economics of security and privacy (2016)

Guest editor, ACM TOIT special issue on pricing and incentives in networks and systems (2013)

TPCs

2027: AISTATS (Senior Area Chair), ICLR (Area Chair)

2026: ICML (Area Chair), NeurIPS (Area Chair), The Web Conf

2025: ICML (Area Chair), NeurIPS (Area Chair)

2024: NeurIPS, ECML-PKDD (Area Chair)

2023: NeurIPS, EWAF (Area Chair), The Web Conf

2022: ICML, ECML-PKDD (Area Chair), The Web Conf, DE

2021: AAAI, FAccT (Area Chair), ECML-PKDD (Area Chair), NetEcon

2020: NeurIPS, ICML, AAAI, IJCAI, PETS, NetEcon

2019: NeurIPS, ICML, SIGMETRICS, PETS, NetEcon

2018: NeurIPS, ICML, NetEcon, NetGCoop, WiOpt, GameNets

2017: NIPS, NetEcon, WiOpt (member of the Advisory TPC), ICPP, ICQT

2016: WWW (demo track), FC, ITC, NetGCooP, SDP

2015: WWW (demo track), SDP

2014: SIGMETRICS, NetGCooP, SDP

2013: W-PIN+NetEcon, SDP, ICQT

2012: GameSec, W-PIN

Invited referee for journals and conferences (each listed only once)

Journal of Machine Learning Research, Operation Research letters, Mathematics of Operation Research, Games and Economics Behavior, Communications of the ACM, Dynamic Games and Applications, Lato Sensu : Revue de la Société de philosophie des sciences, IEEE Open Journal of the Communications Society, Statistics and Probability Letters, Ethics and Information Technology, CDC 2019, WINE 2018, WWW 2018, IEEE Transactions on Signal Processing, ACM Transactions on Privacy and Security, IEEE Transactions on Information Forensics and Security, ACM Transactions on the Web, IEEE Networks, IEEE Transactions on Dependable and Secure Computing, ACM Transactions on Information and System Security, Operation Research, IEEE/ACM Transactions on Networking, International Journal of Information Security, ISAAC 2015, IEEE INFOCOM 2013, Computer Communications journal, IEEE Transactions on Communications, Computer Networks Journal, IEEE Communication Letters, Stochastic Models, ACM SigComm CCR, 20th ITC Specialist Seminar on Network Virtualization - Concepts and Performance, CFIP 2009

Recruitment committees

CoS MCF (Assist. Professor) Univ. Grenoble Alpes (2021)

Evaluation of PhD dissertations

Member of PhD committees: Ziyad Benomar (ENSAE, 2025)—president; Anthony Bardou (Univ. Lyon, 2023)—reviewer; Virginie Do (Univ. Paris Dauphine, 2023)—reviewer; Vincent Grari (Sorbonne Université, 2022)—reviewer; Antoine Rault (Inria Rennes, 2016); Áron Lászka (Budapest University of Technology and Economics, 2014)—reviewer

Member of individual PhD monitoring committees (CSI): Veronika Shilova (Toulouse, 2025), Amal Sakr (Telecom SudParis, 2025), Carlos Pinzòn (Inria Saclay, 2022); Omar Boufous (Université d'Avignon, 2021)

Member of the mid-term evaluation committee for 7 PhD students at EURECOM (2013-17)

Panel member for grant proposal selection

Member of the evaluation committee for the ANR TSIA call on IA (2025)

External expert for the Royal Society (2025)

External expert for the evaluation of startups to enter in incubator Agoranov (2022)

External reviewer for the ANR, France (regularly since 2020)

Expert for the F.R.S.-FNRS, Belgium (regularly since 2016)

Expert for the Czech Science Foundation (2020)

Member of the selection committee for the "Future & Ruptures" program from IMT (2014)

External reviewer for the Informatics and Mathematics Panel of the Academic Research Council, Ministry of Education, Singapore (2014)

Miscellaneous scientific evaluation tasks

Member of the working group for the creation of the Inria team NEO

Awards

Runner-up for the 2019 CNIL-Inria Award for Privacy Protection 2019
(for paper [59] by A. Andreou, G. Venkatadri, O. Goga, K. Gummadi, P. Loiseau and A. Mislove)

Runner-up for Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies 2019
(for paper [57] by G. Venkatadri, A. Andreou, Y. Liu, A. Mislove, K. Gummadi, P. Loiseau and O. Goga)

"Prime d'encadrement doctoral et de recherche (PEDR)" (Inria) 2019-2022

Best Paper Award nominee at FAT* 2018
(for paper [58] by T. Speicher, M. Ali, G. Venkatadri, F. Nunes Ribeiro, G. Arvanitakis, F. Benevenuto, K. P. Gummadi, P. Loiseau, and A. Mislove)

Best Paper Award Runner-up at IEEE/ACM ASONAM 2017
(for paper [61] by A. Andreou, O. Goga, and P. Loiseau)

ANR Tremplin-ERC grant 2016
(Award for researchers who were well evaluated at the second stage of ERC but did not obtain funding)

Humboldt Research Fellowship for experienced researchers (Alexander von Humboldt Foundation) 2016
(Highly competitive; maximum initial award of 18 months, extended in the context of parental support)

Data Transparency lab research grant (top 11% of the projects) 2016
(for a project as co-PI with Oana Goga)

Symantec research faculty gift 2015

Data Transparency lab travel grant (top 30% of the projects) 2015
(for a project as co-PI with Oana Goga)

Best Paper Award Runner-up at IFIP Performance 2010
(for paper [75] by P. Loiseau, P. Gonçalves, J. Barral, and P. Vicat-Blanc Primet)

ERCIM Alain Bensoussan European Post-doctoral fellowship (declined) 2010

Best Student Demonstration Award at ACM SIGMETRICS/Performance 2009
(for demo [93] by P. Loiseau, R. Guillier, O. Goga, M. Imbert, P. Gonçalves, and P. Vicat-Blanc Primet)

PhD fellowship and teaching assistanship from École Normale Supérieure de Lyon 2006

Advising experience

Postdocs

Past:

Denis Sokolov (Dec. 2023–Nov. 2025)

Simon Finster (Nov. 2022–Sept. 2025, co-advised with B. Pradelski, now Assist. Prof at Linz University)

Felipe Garrido Lucero (Oct. 2022–Aug. 2024, co-advised with V. Perchet, now Assist. Prof at Université Toulouse Capitole)

George Arvanitakis (Jan.–Oct. 2018, now researcher at Technology Innovation Institute, Abu Dhabi)

Michela Chessa (Sept. 2013–Aug. 2015, now Assist. Prof at Université Côte d'Azur, GREDEG)

PhD students

Albert Gimó Contreras (Since Jan. 2026, Cifre Criteo, co-advised with M. Vladimirova)

Louise Allain (Since Oct. 2025)

Marie Generali Lince (Since Oct. 2024, co-advised with S. Gaucher)

Minrui Xu (Since Oct. 2024, co-advised with S. Mauras)

Reda Jalal (Since Feb. 2024, co-advised with S. Mauras)

Mélissa Tamine (Since Jan. 2024, Cifre Criteo, co-advised with B. Heymann and M. Vono)

Past:

Mathieu Molina (Jan. 2022–Sept. 2025, co-advised with V. Perchet and N. Gast)
PhD Institut Polytechnique de Paris 2025, now postdoc at Tel-Aviv Univ

Remi Castera (Oct. 2021–Sept. 2024, co-advised with B. Pradelski)
PhD Univ. Grenoble Alpes 2024; now postdoc at UM6P

Till Kletti (Feb. 2020–June 2023, Cifre Naver Labs, co-advised J.-M. Renders and S. Amer-Yahia)
PhD Univ. Grenoble Alpes 2023; now patent officer at the EU Commission

Vitalii Emelianov (Sept. 2018–June 2022, co-advised with N. Gast)
PhD Univ. Grenoble Alpes 2022; now postdoc at University of Udine

Benjamin Roussillon (Oct. 2018–Sept 2021, co-advised with P. Mertikopoulos)
PhD Univ. Grenoble Alpes 2021; now a high-school teacher

Dong Quan Vu (Jan. 2017–June 2020, Cifre Nokia Bell Labs, co-advised with Alonso Silva)
PhD UPMC 2020; now research engineer at Safran Tech

Athanasios Andreou (Oct. 2015–June 2019, co-advised with O. Goga)
PhD UPMC 2019; now postdoc at NYU after a break as an independent consultant

Xiaohu Wu (Nov. 2012–Feb. 2016)
PhD Telecom ParisTech 2016; now Associate Professor at Renmin University (China)

Hadrien Hours (Nov. 2011–Sep. 2015, co-advised with E. Biersack)
PhD Telecom ParisTech 2015; now Machine Learning Scientist at Booking.com

Engineers

Eugénie Patard (Since Nov. 2025)

Interns

Supervision of 25+ MSc (or PhD) level interns

Research funding

EU HORIZON grant 2026 – 2029
AIGENTIC: Building a trustworthy generative AI agents ecosystem for European cyber-resilience and compliance (€ 405k for Inria)
Patrick Loiseau and Oana Goga (co-PIs for Inria partner)

Cifre contract with Criteo 2026 – 2029
Trustworthy AI by Design: Learning with Fairness, Forgetting, and Control Constraints (€ 100k + support of one PhD student)
Patrick Loiseau (co-PI), Mariia Vladimirova (co-PI at Criteo)

Hi! Paris Synergy Fellowship 2025 – 2029
DOLLY: Design, Incentivization, Optimization and (Reinforcement-)Learning of Multi-Layered Market (€ 730k)
Patrick Loiseau (co-PI), Vianney Perchet, Yuki Tamura, and Pablo Winant (co-PIs)

Cifre contract with Criteo 2024 – 2027
Apprentissage coopératif à grande échelle : évaluation de la valeur des données et stratégies d'incitation au partage des données (€ 30k + support of one PhD student)
Patrick Loiseau (co-PI), Maxime Vono, Benjamin Heymann (co-PI at Criteo)

PEPR IA Projet Cible 2023 – 2027
FOUNDRY: FOUNDations of Robustness and reliabilitY in AI (€ 910k for FairPlay)
Patrick Loiseau (PI for FairPlay and contact point for the Inria partners)

ANR JCJC grant 2021 – 2025
FairPlay: Fair algorithms via game theory and sequential learning (€ 245k)
Patrick Loiseau (PI)

Cifre contract with Naver labs 2020 – 2023
Fairness in multi-stakeholder recommendation platforms (€ 75k + support of one PhD student)
Patrick Loiseau (co-PI), Sihem Amer-Yahia (co-PI), Jean-Michel Renders (co-PI at Naver labs)

ANR-NRF France-Singapore PRC grant 2019 – 2023
ALIAS: Adaptive learning for interacting agents and systems (€ 279k for french side)
Bary Pradelski (PI), Patrick Loiseau (member), Panayotis Mertikopoulos (member)

MIAI @ Grenoble Alpes, Chair 2019 – 2023
Explainable and Responsible AI (€ 365k)
Patrick Loiseau (co-PI), Marie-Christine Rousset (co-PI)

Grenoble INP Presidence, doctoral support grant 2019 – 2022
D-TEAM: Dynamic Theory and Experiments for Assignment Markets (€ 50k)
Patrick Loiseau (PI), Bary Pradelski (co-PI), Heinrich Nax (co-PI at Univ of Zurich)

IDEX UGA IRS 2018 – 2021
AdvLearn: Classification in the presence of adversarial data: models and solutions (€ 50k)
Patrick Loiseau (PI), Panayotis Mertikopoulos (co-PI)

DGA, doctoral grant 2018 – 2021
Classification en présence de données adverses : modèles et solutions (€ 59k)
Patrick Loiseau (PI), Panayotis Mertikopoulos (co-PI)

ANR Tremplin-ERC 2017 – 2019
CONNECTED: Towards secure and private personal-data-based online services in the networked world (€ 150k)
Patrick Loiseau (PI)

Cifre contract with Nokia Bell Labs 2016 – 2019
Learning in Blotto Games and Applications to Modeling Attention in Social Networks (€ 45k + support of one PhD student)
Patrick Loiseau (co-PI), Alonso Silva (co-PI at Nokia Bell Labs)

Cifre contract with SAP Research 2016 – 2019
Approche de l'anonymisation des données en fonction du niveau de risque associé (€ 45k + support of one PhD student)
Patrick Loiseau (co-PI), Michele Bezzi (co-PI at SAP Research)

Data Transparency lab research grant 2016 – 2017
TranspAd: A Collaborative Tool to Bring Transparency to Targeted Advertising (€ 50k)
Patrick Loiseau (co-PI), Oana Goga (co-PI)

Institut Mines-Telecom Futur&Ruptures program, doctoral support grant 2015 – 2018
TRANSPA: Bringing transparency to personalized services through statistical inference (€ 108k)
Patrick Loiseau (PI)

France-Berkeley fund 2014 – 2016
Multi-armed bandit games and applications ($ 10k)
Patrick Loiseau (PI), Jean Walrand (PI)

Symantec research faculty gift 2015
Cyber insurance ($ 30k)
Patrick Loiseau (PI)

Institut Mines-Telecom Futur&Ruptures program, post-doctoral support grant 2015
MONET: MONETization of personal data in social networks: A game-theoretic approach (€ 30k)
Patrick Loiseau (PI)

Labex UCN@Sophia, post-doctoral support grant 2013 – 2015
PRIMO: PRIvate data MOnetization: a public good approach using cooperative game theory (€ 90k)
Patrick Loiseau (PI)

Labex UCN@Sophia, doctoral support grant 2013 – 2016
Mathematical tools for the smart grid (€ 105k)
Patrick Loiseau (co-PI), Giovanni Neglia (co-PI)

Institut Mines-Telecom Futur&Ruptures program, doctoral support grant 2012 – 2015
Robust pricing of cloud resources through mean-field games (€ 121k)
Patrick Loiseau (PI)

Publications

My publications are available at `https://patrickloiseau.github.io/publications.html`.
Link to Google Scholar profile: `https://scholar.google.fr/citations?user=q98gB0AAAAAJ&hl=en`.

Preprints
(Excluding papers submitted to conferences with anonymous submissions.)

[1] Felipe Garrido-Lucero, Denis Sokolov, **Patrick Loiseau**, and Simon Mauras. Two-sided matching with resource-regional caps, 2025. (Preprint available at `https://arxiv.org/pdf/2502.14690`).

[2] Mohamed Alzayat, Johnnatan Messias, Balakrishnan Chandrasekaran, Krishna P. Gummadi, and **Patrick Loiseau**. Modeling Coordinated vs. P2P Mining: An Analysis of Inefficiency and Inequality in Proof-of-Work Blockchains, 2021. (Preprint available at `https://arxiv.org/abs/2106.02970`).

[3] Dong Quan Vu, **Patrick Loiseau**, and Alonso Silva. Approximate equilibria in non-constant-sum colonel blotto and lottery blotto games with large numbers of battlefields, 2020. (Preprint available at `https://arxiv.org/abs/1910.06559`).

[4] Michela Chessa and **Patrick Loiseau**. A cooperative game-theoretic approach to quantify the value of personal data in networks, 2018. (Preprint available at `https://ideas.repec.org/p/gre/wpaper/2018-02.html`).

Articles in journals

[5] Rémi Castera, **Patrick Loiseau**, and Bary S.R. Pradelski. Correlation of rankings in matching markets. *Management Science*, 72(9):7591–7607, 2026.

[6] Michela Chessa and **Patrick Loiseau**. Enhancing voluntary contributions in a public goods economy via a minimum individual contribution level. *Public Choice*, 201:237–261, April 2024.

[7] Xiaohu Wu and **Patrick Loiseau**. Algorithms for scheduling deadline-sensitive malleable tasks. *Operations Research Forum*, 5(30), 2024.

[8] Vijay Kamble, **Patrick Loiseau**, and Jean Walrand. An approximate dynamic programming approach to adversarial online learning. *Operations Research*, 72(1):373–388, 2024.

[9] Xiaohu Wu and **Patrick Loiseau**. Efficient approximation algorithms for scheduling moldable tasks. *European Journal of Operations Research*, 310(1):71–83, October 2023.

[10] Vitalii Emelianov, Nicolas Gast, Krishna P. Gummadi, and **Patrick Loiseau**. On fair selection in the presence of implicit and differential variance. *Artificial Intelligence*, 302:103609, January 2022.

[11] Nicolas Gast, Stratis Ioannidis, **Patrick Loiseau**, and Benjamin Roussillon. Linear regression from strategic data sources. *ACM Transactions on Economics and Computation*, 8(2):10:1–10:24, May 2020.

[12] Xiaohu Wu, **Patrick Loiseau**, and Esa Hyytia. Towards designing cost-optimal policies to utilize iaas clouds with online learning. *IEEE Transactions on Parallel and Distributed Systems*, 31(3):501–514, March 2020.

[13] Alberto Benegiamo, **Patrick Loiseau**, and Giovanni Neglia. Dissecting demand response mechanisms: The role of consumption forecasts and personalized offers. *Sustainable Energy, Grids and Networks*, 16:156–166, December 2018.

[14] Lemonia Dritsoula, **Patrick Loiseau**, and John Musacchio. A game-theoretic analysis of adversarial classification. *IEEE Transactions on Information Forensics and Security*, 12(12):3094–3109, December 2017.

[15] Raimo Kantola, Hammad Kabir, and **Patrick Loiseau**. Cooperation and End-to-End in the Internet. *International Journal of Communication Systems*, 30(12):e3268, August 2017.

[16] Hadrien Hours, Ernst Biersack, **Patrick Loiseau**, Alessandro Finamore, and Marco Mellia. A Study of the Impact of DNS Resolvers on CDN Performance Using a Causal Approach. *Computer Networks, Special issue on "Traffic and Performance in the Big Data Era"*, 109, Part 2:200 – 210, 2016.

[17] Hadrien Hours, Ernst Biersack, and **Patrick Loiseau**. A causal approach to the study of TCP performance. *ACM Transactions on Intelligent Systems and Technology, Special Issue on "Causal Discovery and Inference" (K. Zhang, J. Li, E. Bareinboim, B. Schölkopf, and J. Pearl, editors)*, 7(2):25:1–25:25, January 2016.

[18] **Patrick Loiseau**, Galina Schwartz, John Musacchio, Saurabh Amin, and S. Shankar Sastry. Incentive mechanisms for internet congestion management: Fixed-budget rebate *versus* time-of-day pricing. *IEEE/ACM Transactions on Networking*, 22(2):647–661, 2014.

[19] **Patrick Loiseau**, Claire Médigue, Paulo Gonçalves, Najmeddine Attia, Stéphane Seuret, François Cottin, Denis Chemla, Michel Sorine, and Julien Barral. Large deviations estimates for the multiscale analysis of heart rate variability. *Physica A*, 391(22):5658–5671, November 2012.

[20] Paulo Gonçalves, Shubhabrata Roy, Thomas Begin, and **Patrick Loiseau**. Dynamic resource management in clouds: A probabilistic approach. *IEICE Transactions on Communications, special section on Networking Technologies for Cloud Services*, E95-B(8):2522–2529, 2012. (Invited paper).

[21] Julien Barral and **Patrick Loiseau**. Large deviations for the local fluctuations of random walks. *Stochastic Processes and their Applications*, 121(10):2272–2302, 2011.

[22] **Patrick Loiseau**, Paulo Gonçalves, and Pascale Vicat-Blanc Primet. A long-range dependent model for network traffic with flow-scale correlations. *Stochastic Models*, 27:333–361, 2011.

[23] Edmundo Pereira de Souza Neto, Elmer Andrés Fernández, Patrice Abry, Béatrice Cuzine, **Patrick Loiseau**, Christian Baude, Jean Frutoso, Claude Gharib, and Xavier Martin. Application of cardiac autonomous indices in the study of neurogenic erectile dysfunction. *Urologia Internationalis*, 86(3):290–297, 2011.

[24] **Patrick Loiseau**, Paulo Gonçalves, Guillaume Dewaele, Pierre Borgnat, Patrice Abry, and Pascale Vicat-Blanc Primet. Investigating self-similarity and heavy-tailed distributions on a large scale experimental facility. *IEEE/ACM Transactions on Networking*, 18(4):1261–1274, August 2010.

[25] Edmundo Pereira de Souza Neto, Patrice Abry, **Patrick Loiseau**, Jean-Christophe Cejka, Marc-Antoine Custaud, Jean Frutoso, Claude Gharib, and Patrick Flandrin. Empirical mode decomposition to assess cardiovascular autonomic control in rats. *Fundamental & Clinical Pharmacology*, 21(5):481–496, October 2007.

Articles in international refereed conferences

[26] Albert Gimó Contreras, Mariia Vladimirova, Olga Petrova, Reda Chhaibi, and **Patrick Loiseau**. Scalable fair learning via cramér-von mises regularization. In *Proceedings of the 40th Conference on Neural Information Processing Systems (NeurIPS)*, December 2026.

[27] Marie Generali Lince, Vincent Divol, Rémi Flamary, Solenne Gaucher, and **Patrick Loiseau**. Geometry of relaxed fair regression: A unified framework for aware and unaware settings. In *Proceedings of the 40th Conference on Neural Information Processing Systems (NeurIPS)*, December 2026.

[28] Eugénie Patard, Corentin Pla, Benjamin Heymann, and **Patrick Loiseau**. Dynamauction: a reinforcement learning environment for repeated auction with dynamic value. In *Proceedings of the 40th Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Datasets track*, December 2026.

[29] **Patrick Loiseau**, Simon Mauras, and Minrui Xu. Complexity of auctions with interdependence. In *Proceedings of the 27th ACM Conference on Economics and Computation (EC)*, July 2026.

[30] Mélissa Tamine, Benjamin Heymann, Maxime Vono, and **Patrick Loiseau**. On the impact of the utility in semivalue-based data valuation. In *Proceedings of the the Fourteenth International Conference on Learning Representations (ICLR)*, May 2026.

[31] Rémi Castera, Felipe Garrido Lucero, **Patrick Loiseau**, Simon Mauras, Mathieu Molina, and Vianney Perchet. The price of opportunity fairness in matroid allocation problems. In *Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS)*, December 2025.

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[79] Hadrien Hours, Ernst Biersack, and **Patrick Loiseau**. Causal study of network performance. In *Proceedings of the 17ème Rencontres Francophones sur les Aspects Algorithmiques de Télécommunications (AlgoTel)*, June 2014.

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[81] **Patrick Loiseau**, Paulo Gonçalves, and Pascale Vicat-Blanc Primet. How TCP can kill self-similarity. In *Euro-NF workshop: Traffic Engineering and Dependability in the Network of the Future*, September 2008.

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[83] **Patrick Loiseau**, Paulo Gonçalves, Guillaume Dewaele, Pierre Borgnat, Patrice Abry, and Pascale Vicat-Blanc Primet. Vérification du lien entre auto-similarité et distributions à queues lourdes sur un dispositif grande échelle. In *9ème Atelier en Evaluation de Performances (AEP9)*, June 2008.

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[85] Nicolas Gast, Stratis Ioannidis, **Patrick Loiseau**, and Benjamin Roussillon. Linear regression from strategic data sources. In *the ACM EC 2019 Workshop on Learning in Presence of Strategic Behavior*, June 2019.

[86] Dong Quan Vu, **Patrick Loiseau**, and Alonso Silva. Approximate equilibria of the Colonel Blotto game. In *the 14th European meeting on game theory (SING14)*, July 2018. (1-page abstract).

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[88] Michela Chessa, Jens Grossklags, and **Patrick Loiseau**. On non-monetary incentives for the provision of public goods. In *the 13th European meeting on game theory (SING13)*, July 2017. (1-page abstract).

[89] Michela Chessa, Jens Grossklags, and **Patrick Loiseau**. A game-theoretic study on non-monetary incentives in data analytics projects with privacy implications. In *8e édition de l'Atelier sur la Protection de la Vie Privée (APVP)*, June 2017.

[90] Vijay Kamble, **Patrick Loiseau**, and Jean Walrand. Regret-optimal strategies for playing discounted repeated games. In *27th International Conference on Game Theory in Stony Brook*, July 2016. (1-page abstract).

[91] Michela Chessa and **Patrick Loiseau**. The impact of the graph structure on a public good provision game: a cooperative approach with applications to personal data release on social networks. In *the 11th European meeting on game theory (SING11-GTM2015)*, July 2015. (1-page abstract).

[92] **Patrick Loiseau**, Paulo Gonçalves, Romaric Guillier, Matthieu Imbert, Oana Goga, Yuetsu Kodama, and Pascale Vicat-Blanc Primet. *Metroflux*: a high performance system for very fine-grain flow analysis. In *Grid'5000 Spring School*, April 2009.

Software Demonstrations

[93] **Patrick Loiseau**, Romaric Guillier, Oana Goga, Matthieu Imbert, Paulo Gonçalves, and Pascale Vicat-Blanc Primet. Automated traffic measurements and analysis in Grid5000, June 2009. ACM SIGMETRICS / Performance demonstration contest (**Best Student Demonstration Award**).

Edited volumes

[94] Rainer Böhme, Richard Clayton, Jens Grossklags, Katrina Ligett, **Patrick Loiseau**, and Galina Schwartz. Special Issue on the Economics of Security and Privacy: Guest Editors' Introduction. *ACM Transactions on Internet Technology*, 18(4):47:1–47:3, November 2018. (Guest editorial).

[95] **Patrick Loiseau**, Aaron Roth, and Adam Wierman. The 10th Workshop on the Economics of Networks, Systems and Computation (NetEcon 2015). *ACM Performance Evaluation Review*, 43(3):47–48, December 2015. (Guest editorial).

[96] John Chuang and **Patrick Loiseau**. The joint Workshop on Pricing and Incentives in Networks and Systems (W-PIN+NetEcon 2014). *ACM Performance Evaluation Review*, 42(3):2–3, December 2014. (Guest editorial).

[97] Costas Courcoubetis, Roch Guérin, **Patrick Loiseau**, David Parkes, Jean Walrand, and Adam Wierman. Special Issue on Pricing and Incentives in Networks and Systems: Guest Editors' Introduction. *ACM Transactions on Internet Technology*, 14(2–3):8:1–8:3, October 2014. (Guest editorial).

[98] **Patrick Loiseau**, David Parkes, and Jean Walrand. The joint Workshop on Pricing and Incentives in Networks and Systems (W-PIN+NetEcon 2013). *ACM Performance Evaluation Review*, 41(4):2–3, March 2014. (Guest editorial).

[99] **Patrick Loiseau** and Jean Walrand. The first Workshop on Pricing and Incentives in Networks (W-PIN 2012). *ACM Performance Evaluation Review*, 40(2):12–13, September 2012. (Guest editorial).

Dissertations

[100] **Patrick Loiseau**. *Combining game theory and statistical learning for security, privacy and network systems*. HDR thesis, UPMC, December 2016.

[101] **Patrick Loiseau**. *Contributions to the Analysis of Scaling Laws and Quality of Service in Networks: Experimental and Theoretical Aspects*. PhD thesis, ENS Lyon, December 2009.

Software

AdAnalyst: A Chrome and Firefox extension that provides aggregate statistics and various insights and visualizations about the ads a user receives on Facebook.
Available at `http://adanalyst.mpi-sws.org/`.

Invited talks / Keynotes / Guest lectures

Keynotes and invited lectures

Keynote Netgcoop conference (Bilbao, 2025); Tutorial Hi! Paris Summer School (Paris, 2023); Keynote RAWNET workshop @ WiOpt (Avignon, 2019); Lecture Data science summer school DS3 (Paris, 2019); Plenary talk MSTIC days (Grenoble, 2017); Keynote 11th Workshop on Performance Evaluation (Toulouse, 2016); Lecture IPAM Grad. Summer School on Games and Contracts for Cyber-Physical Security (UCLA, 2015); Tutorial ACM SIGMETRICS (Portland, 2015); Lecture RESCOM summer school (Vitel, 2012)

Invited talks at conferences/workshops and seminars

Crossroads of AI & Society (Paris, 2026); GAIMSS workshop (Toulouse, 2025); PEPR day @ COLT (Lyon, 2025); Workshop From Matching to Market. A Tale of Mathematics, Economics and Computer Science (Cargese, 2025); Ethics of Public Robots and AI (Paris, 2023); European Meeting of Statisticians (Varsaw, 2023); LSE data science semainar (London, 2023); Columbia CS seminar (New-York City, 2023); Workshop @ Comete on Ethical AI (Palaiseau, 2022); Atelier pluridisciplinaire sur la justice sociale (Cachan, 2021); Columbia CS seminar (New-York City, 2018); SIESTE seminar (ENS Lyon, 2016), MPI-SWS seminar (Saarbrücken, 2016); Harvard EconCS seminar (Cambridge, 2015); MIT Special Henry L. Pierce seminar (Cambridge, 2015); Northeastern University ECE department seminar (Boston, 2015); Data Transparency Lab conference @ MIT (Cambridge, 2015); LINCS seminar (Paris, 2015); Paris game theory seminar @ IHP (Paris, 2015); Séminaire d'algorithmique et de complexité du plateau de Saclay (Saclay, 2015); In'tech seminar (Grenoble, 2015); Data transparency lab kickoff workshop (Barcelona, 2014); AlgoGT (Saint Nizier du Moucherotte, 2013); SophiaTech networks seminar (Sophia-Antipolis, 2013); Mines ParisTech CMA seminar (Sophia-Antipolis, 2013); TRUST seminar (Berkeley, 2013); Supélec seminar (Gif-sur-Yvette, 2012); Inria RAP seminar (Paris, 2012); UCLA EE dept seminar (Los Angeles, 2011); Caltech RSRG Seminars (Pasadena, 2011); Orange Labs seminar (Sophia-Antipolis, 2011); Séminaire de Probabilités et Statistiques (Nice, 2011); Alcatel-Lucent Bell Labs Mathematics of Networks and Communications Research Department seminar (Murray Hill, 2011); University of Waterloo Department of Electrical and Computer Engineering invited seminar (Waterloo, 2010); Networking, Communications and DSP seminars (Berkeley, 2010); Caltech RSRG Seminars (Pasadena, 2010); Politecnico di Torino Telecommunication Network seminar (Torino, 2009)

Dissemination of scientific knowledge

Articles

Oana Goga and Patrick Loiseau. *Publicité en Ligne : reprenons la main !* Dissemination article co-published by the blog Binaire (Le Monde) and The Conversation France, June 3, 2019.

Conferences

Algorithmes et société: transparence de la publicité en ligne. Mini-conférence grand public, Festival Transfo, Grenoble, France, January 2018.

Misc

Participation in a debate "C'est pas moi, c'est l'IA", Biennale Arts-Sciences EXPERIMENTA, February 2020.

Organization of a workshop *IA, éthique et société*, Forum Ecobiz Grenoble, October 2019.

Participation in a debate *Ethique et numérique : quels enjeux sociétaux ?.* Festival Transfo, Grenoble, France, January 2019.

Press coverage

Tout ce que Facebook sait de moi (et de vous), Le Figaro, April 10, 2018.

"Why am I seeing this ad" explanations on Facebook are incomplete and misleading, a study says, Quartz, April 06, 2018.

Does Facebook's Ad Tool Mislead Voters? Bloomberg Businessweek, March 26, 2018.

To understand digital advertising, study its algorithms. The Economist, March 22, 2018.

'It might work too well': the dark art of political advertising online. The Guardian, March 19, 2018.

Is Facebook being honest with you about how it targets ads? Australian Broadcasting Corporation, March 18, 2018.

We still have no idea what really happens on Facebook. The Verge, March 15, 2018.

Researchers Discovered Data Leak In Facebook's Ad Software. Fast Company, March 6, 2018.

Facebook bug could have exposed your phone number to marketers. Naked Security, Jan. 9, 2018.

Facebook bug let advertisers find users' phone numbers. The Telegraph, Jan. 9, 2018.

Une faille dans Facebook laissait les annonceurs récupérer nos numéros de téléphone. Siècle digital, Jan. 8, 2018.

Facebook Bug Could Let Advertisers Get Your Phone Number. WIRED, Jan. 7, 2018.

Algorithms Are Getting Better at Matching Your Different Social Media Profiles. But they're still pretty lousy at it. MOTHERBOARD, June 10, 2015.